

When the Earth Moves

Earthquakes in Peru

Starters edition · Pre-A1



Before you start

BIG QUESTION

Peru shakes. What do I have to know, and what do I have to have ready, so that it does not catch my family by surprise?

Peru is a big country. It has a long coast, high mountains and a big forest.

Sometimes the ground in Peru moves. We call it an **earthquake**.

In this book you read about real earthquakes in Peru. You learn what to do. Then you make a **safety flyer** for a family in your street.

Meet The Lighthouse Explorers



Freya

shows you the new words



Nico

makes mistakes — can you help him?



Astrid

gives the instructions



Tomás

loves action and never stops



Pip

is hiding in every chapter.
Can you find him?



Find Pip! Pip the puffin is hiding at the start of every chapter. Can you find him every time?

Contents

1	Our Ground Moves	W1
2	Reading the Numbers	W2
3	What Happened in 1970	W3
4	When the Sea Comes	W4
5	Mountains of Fire	Science link
6	When People Get Hurt	W5
7	The Backpack	W6
8	The Safe Zone and the Drill	W7
9	How a Safety Flyer Works	W8
10	Writing My Flyer	W9
11	Making It Better	W10
12	The Safety Fair	W11
•	Peru in Photos	
•	Timeline of Peru's Earthquakes	
•	Timeline of the World	
•	Glossary	

- **How am I doing? (Rubric)**

.....

- **Sources and Credits**

.....

Our Ground Moves



Monday morning at school.



What's this? The floor is moving!

It is Monday morning at school in Lima. The children are reading.

Then the windows shake. The lamp moves. The floor moves too!

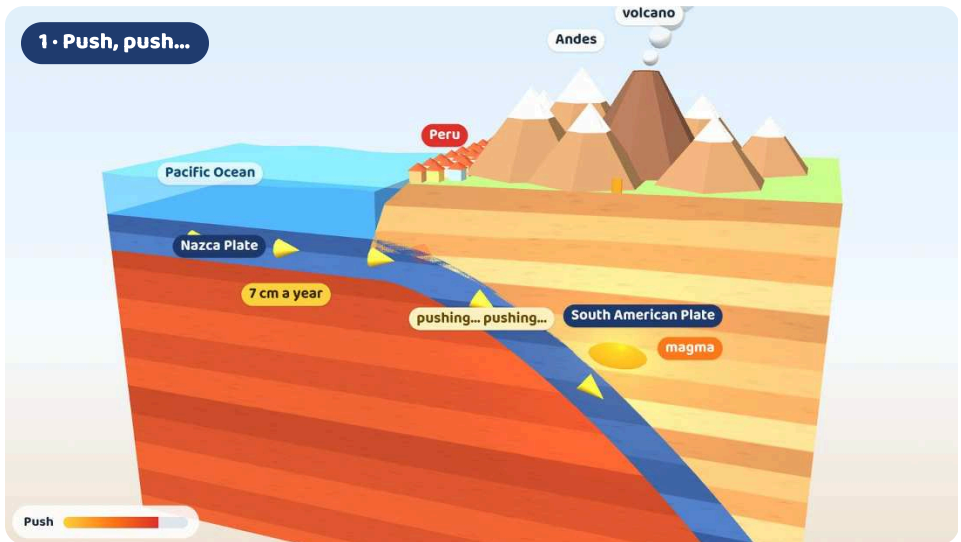
“What's that?” asks Nico.

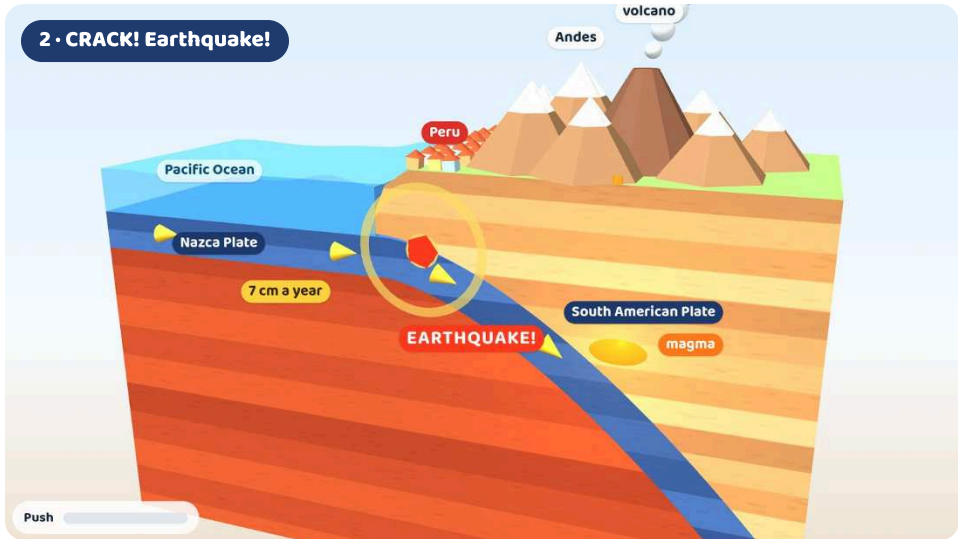
“It's a small earthquake,” says the teacher. “Stay calm.” After ten seconds, the ground stops. Everybody is OK.

A giant puzzle

The Earth is like a big puzzle. The pieces are very, very big. They are called **plates**.

Peru is on a plate. The sea next to Peru is on a plate too. It is called the **Nazca Plate**.



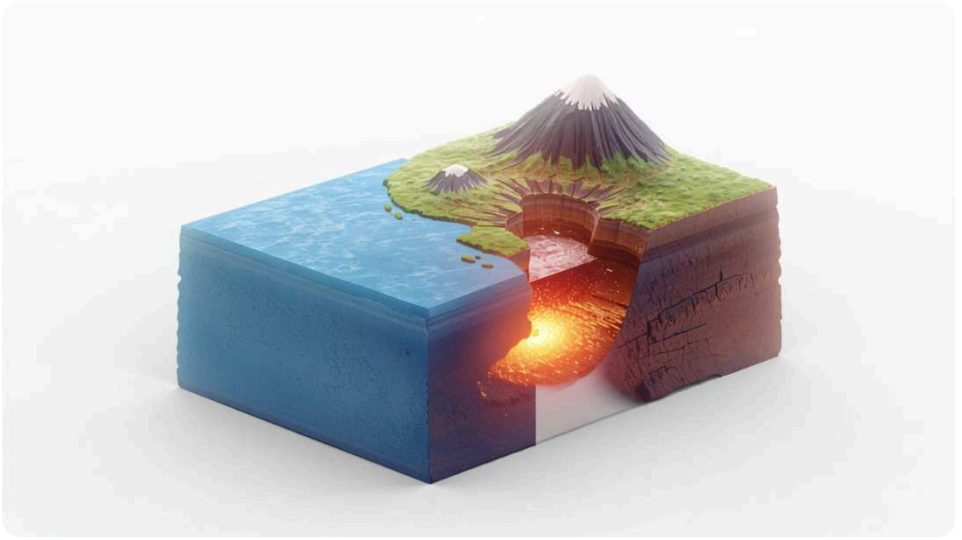


The Nazca Plate goes under Peru. It pushes... then CRACK! An earthquake.

3D model — watch it move in the web edition.

The Nazca Plate goes under Peru. It moves a little every year.

The plates push and push. Then — crack! — they move. The ground shakes. That is an earthquake.



Under the sea, one plate goes under the other.

The Ring of Fire

Look at the map. There is a big ring around the Pacific Ocean. It is the **Ring of Fire**. Peru is on the Ring of Fire. There are a lot of earthquakes and volcanoes there.



The Ring of Fire. Find Peru!

3D model — interactive in the web edition.

The Puzzle Man

Alfred Wegener was a scientist from Germany. In 1912 he looked at a map. He saw that South America and Africa fit together, like two puzzle pieces! He said: “The land moves.” Many people said, “No!” But he was right.



Photo: Unknown photographer · Public domain

Two every day

In Peru there are a lot of earthquakes. In 2025 there were more than 800! That is two every day. Most of them are very small.



LOOK!

Nobody can say when an earthquake will come. So we must be ready!

The words for what the Earth does



earthquake

The ground shakes.



tsunami

A very big wave from the sea.



landslide

Rocks and mud fall down a hill.



flood

Water covers the streets.

Animals and Earthquakes

Some people say animals know before an earthquake. Dogs bark. Chickens run. But scientists say: "We don't know!"



Now you try!

Chapter 1 · Our Ground Moves

A Draw the four words. Hand in · W1

Draw a picture for each word. Then copy the word.

earthquake

tsunami

landslide

flood

B Match the words and the meanings.

1. earthquake

a. a big wave

2. tsunami

b. water on the streets

3. landslide

c. the ground shakes

4. flood

d. rocks fall down a hill

C Yes or no?

1. Peru is on the Ring of Fire. **yes / no**
2. The Earth is one big piece. **yes / no**
3. In Peru there are earthquakes every day. **yes / no**
4. We know when the next earthquake comes. **yes / no**

D Look at the picture of the plates. Write the words.

Words: sea • Peru • plate • Andes

1 _____ 2 _____ 3 _____ 4 _____

E Have you felt one? Ask five friends.

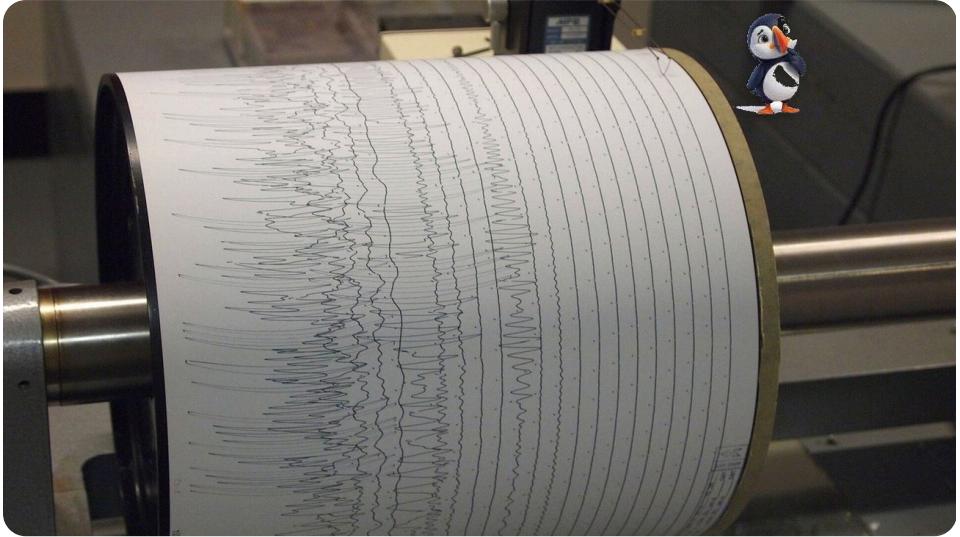
Ask: “Do you feel earthquakes?” Tick ✓ yes or ✗ no.

Name	Yes / No
_____	_____
_____	_____
_____	_____
_____	_____

F Count!

Two earthquakes every day. How many in one week? $2 + 2 + 2 + 2 + 2 + 2 + 2$
= _____

Reading the Numbers



A seismograph draws the shaking of the ground as a zig-zag line.

Photo: Z22 · CC BY-SA 3.0



Look at the table. Read the numbers with me!

Scientists have a special machine. It is a **seismograph**.
When the ground shakes, the machine draws a line. A big
line is a big earthquake.

Richter Scale

Charles Richter was a scientist in the USA. In 1935 he made numbers for earthquakes. A 7 is very strong. An 8 is stronger. A 9 is the strongest!



Photo: Gil Cooper, Los Angeles Times · CC BY 4.0

Magnitude or Intensity?

Magnitude is **one** number for the whole earthquake. **Intensity** is how much you feel it where you are. You feel it more near the **epicentre**. After a big earthquake, small earthquakes can come. They are **aftershocks**.

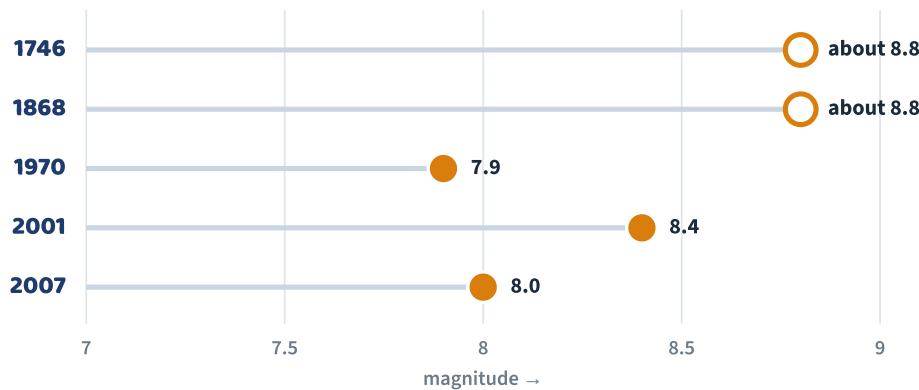
Five earthquakes Peru remembers

Peru has earthquakes every day. But five were very, very big. Look at the table.

Five earthquakes Peru remembers

Year	Where?	Magnitude	Tsunami?
1746	Lima and Callao	about 8.8	yes 
1868	Arica	about 8.8	yes 
1970	Áncash	7.9	no
2001	Arequipa, Moquegua, Tacna	8.4	yes 
2007	Pisco, Ica, Chincha	8.0	small

Which is the biggest number? Which earthquake has a tsunami?



Magnitude of the five earthquakes. The big numbers are the strongest.

The IGP: Peru's Earthquake Watchers

The IGP is in Peru. The scientists watch the ground every day and every night.

**LOOK!**

Magnitude 9.5 — Chile, 1960 — is the biggest earthquake ever measured.

Asking about data

Ask your friend: **When was the big earthquake in Pisco?**

— “In 2007.”

Now you try!

Chapter 2 · Reading the Numbers

A Copy five true sentences. Hand in · W2

Circle the true sentences. Copy them.

1. The Pisco earthquake was in 2007.
 2. The Áncash earthquake was in 1970.
 3. There was a tsunami in Arica.
 4. The 2001 earthquake was in Lima.
 5. The 1746 earthquake was in Lima and Callao.
 6. There was a big tsunami in 1970.
-
-
-

B Read the table and answer.

1. When was the Pisco earthquake? _____
2. Where was the 1868 earthquake? _____
3. How many earthquakes are in the table? _____

C Big, bigger...

8.8 is **bigger** than 7.9. ✓ 7.9 is bigger than 8.4. ✗

Now you: 8.4 is _____ than 8.0.

D Read the chart.

Look at the magnitude chart. Colour the strongest earthquakes red.

E Ten times!

A magnitude 5 makes waves 10 times bigger than a 4. Draw a small wave and a big wave.

F Speaking: data detectives

Point to the table. Say: “1970. Áncash. 7.9.” Your friend says the right row.

What Happened in 1970



The big mountain Huascarán and the town of Yungay.



What happened in Yungay? Let's read.

It is Sunday, 31 May 1970. It is a sunny day in Áncash. There are big mountains. The highest mountain is **Huascarán**. It has a lot of snow.

In the town of Yungay, people are happy. Some families watch football on TV. Children go to the circus.

At 3:23 in the afternoon, the ground shakes. It shakes for 45 seconds. Many houses fall.



Ice and rocks fall from the mountain.

Then, on the mountain, a very big piece of ice and rock falls. It goes down very, very fast. It covers the town of Yungay with mud and rocks.

Some people run up a hill. It is the cemetery. They are safe. The children at the circus are safe too. But many, many people die. It is very, very sad.

The Four Palm Trees

After the avalanche, you can see the tops of four palm trees. They are still there today!



Photo: Yhhue91 · CC BY-SA 4.0



Yungay today. It is a quiet, green park.

After the earthquake

People from all of Peru help. People from other countries help too. They bring food, water and tents.



Astrid's tip

It is 1970. We can tell a story in the present: *The ground shakes. The ice falls. People run.*

Now you try!

Chapter 3 · What Happened in 1970

A Your report about 1970 — four sentences.

 Hand in · W3

Complete and copy.

1. It is Sunday, 31 May _____.
 2. The ground _____.
 3. Ice and rocks _____ from Huascarán.
 4. People _____ up the hill.
-
-

B Put the events in order (1–4).

- ☐ Ice and rocks fall.
- ☐ It is a sunny Sunday.
- ☐ The ground shakes.
- ☐ People go up the hill.

C Find the words.

Circle: **mountain · ice · pizza · town · snow · car** — Which words are in the story?

D Then and now

Look at the two pictures of Yungay. Say one thing: “Now there is a park.”



A message of solidarity

Draw a heart and write: **We remember Yungay.**

When the Sea Comes



A tsunami is a very big wave.



Oops! Is a tsunami one wave? No — many waves!

It is night. It is the year 1746. The ground in Lima shakes. Then the sea comes. A very big wave covers the port of Callao.

What is a tsunami?

An earthquake under the sea moves the water. The water makes big waves. They are called a **tsunami**. The waves are very fast.

A Japanese Word

“Tsunami” is a Japanese word. It means “harbour wave”.

The ship in the desert

In 1868, in Arica, a big tsunami comes. It carries a ship onto the land! The ship is on the sand, far from the sea.



A ship on the sand!

The earthquake is the alarm

Are you near the sea? Does the ground shake a lot? Go up!
Go to a high place. Don't wait.

Five steps

- 1  The ground shakes a lot. It is your alarm.
- 2  Walk. Go away from the beach.
- 3  Go up! Go to a high place.
- 4  Stay there. More waves come.
- 5  Go back when adults say “OK”.

A Wave from Far Away

In 2025 there was a big earthquake in Russia. Peru closed the beaches. The waves were small.



Photo: Gaboct - CC BY-SA 4.0

Now you try!

Chapter 4 · When the Sea Comes

A The five steps — in order. Hand in · W4

Number the pictures and sentences 1–5. Copy them in your notebook.

- ☐ Stay there.
- ☐ The ground shakes a lot.
- ☐ Go up!
- ☐ Go back when adults say OK.
- ☐ Walk away from the beach.

B Tsunami: true or false?

1. A tsunami is a big wave. **yes / no**
2. We go to the beach to look. **yes / no**
3. We go to a high place. **yes / no**

C Read the signs.

Look at the green sign with the arrow. What does it say? Colour an arrow green.

D The ship in the desert — questions

What is on the sand? A _____.

E Role-play: on the beach

Play in pairs. One says: “The ground shakes!” The other says: “Go up!” and walks up the stairs.

CHAPTER 5

Science link · Reading plan

Mountains of Fire

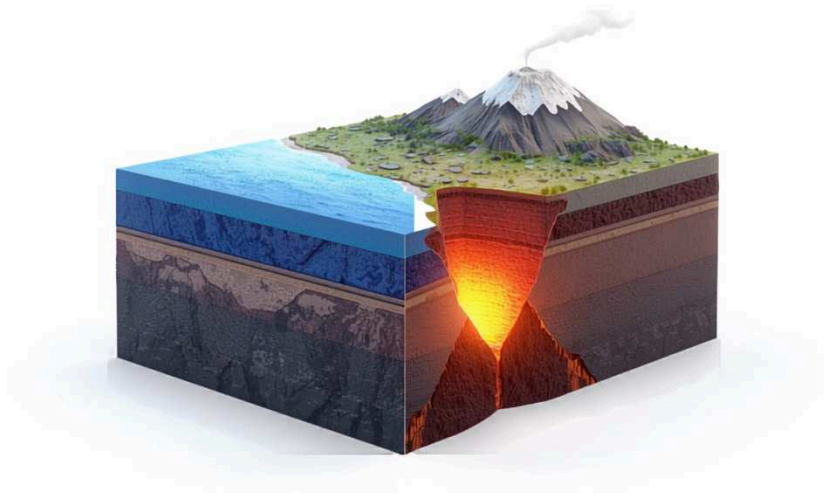


The volcano El Misti and the city of Arequipa.



What's this? A mountain with smoke!

A **volcano** is a mountain. Inside, there is very hot rock. It is called **lava**. Sometimes smoke and ash come out of the top.



Hot rock goes up inside the volcano.

Peru has a lot of volcanoes. They are in the south of Peru.

The biggest eruption in South America

In the year 1600, a volcano in Moquegua explodes. Its name is **Huaynaputina**. There is a lot of ash. The ash covers the city of Arequipa.



A volcano erupts. The llamas run away!

The White City

Arequipa is the “White City”. Many houses are made of white volcano stone!



Photo: Josep M. Gracia · CC BY-SA 4.0

Some volcanoes in Peru are active today. **Sabancaya** has ash every day!



LOOK!

A volcano can be higher than 5,000 metres!

Now you try!

Chapter 5 · Mountains of Fire

A Label the volcano.

Draw a volcano. Write: **top** · **smoke** · **lava**

B Big, bigger, biggest

Sabancaya 5,960 m · El Misti 5,822 m · Ubinas 5,672 m

The highest volcano is _____.

C True or false?

1. Peru has volcanoes. **yes / no**
2. The volcanoes are in the north. **yes / no**

D Make a volcano! (with your teacher)

You need: a bottle, baking soda, vinegar, red colour. Put in the soda. Pour in the vinegar. Look!

CHAPTER 6

W5 · Final y entrega

When People Get Hurt



Pisco, 2007. Many houses fall.

Photo: Senior Airman Shaun Emery, U.S. Air Force · Public domain (U.S. military)



My knee hurts! Who can help me?

It is 15 August 2007. It is 6:40 in the evening. In Pisco, the ground shakes for a long time. Many houses fall. Many people are hurt.

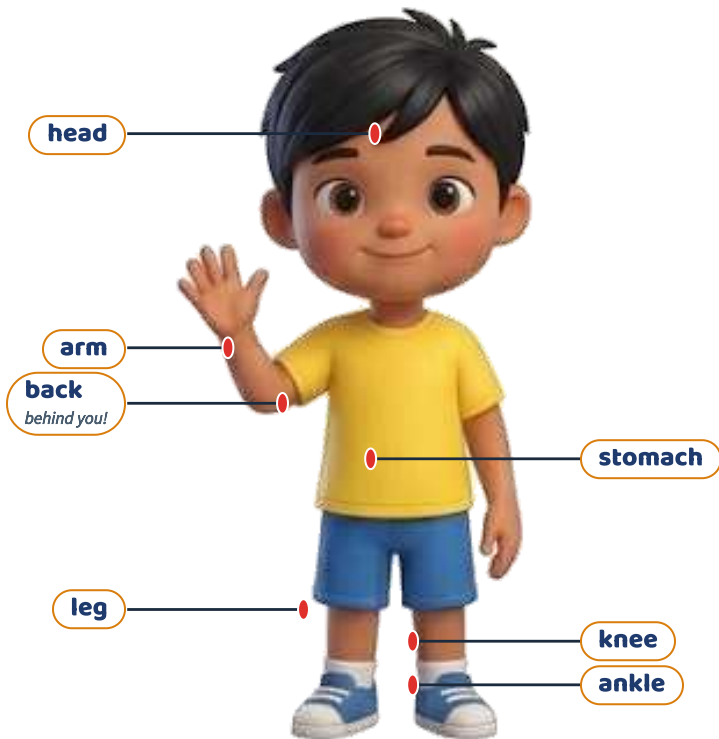
Firefighters help. Doctors help. People from all of Peru bring food and water.



Planes bring help.

Photo: Tech. Sgt. Jeremy Lock, U.S. Air Force · Public domain (U.S. military)

My body



Point to your head, arm, back, leg, knee and ankle!

Where does it hurt? — My arm hurts. · My leg hurts. · My head hurts.

I have a headache. (head + ache)

Who do you tell? What do you say?

Tell an adult: a teacher, your mum or your dad. Say:
“Help, please! My friend is hurt.”



116 Firefighters
Bomberos



105 Police
Policia



106 Ambulance
SAMU



115 Civil defence
Defensa Civil



119 Leave a voice message
Mensajes de voz

Line 119: Leave a Message

After a big earthquake, phones can be busy. Call 119 and leave a message for your family!



Freya: Are you OK?



Nico: No. My arm hurts.



Freya: Sit here. I can call the teacher.



Nico: Thank you!

Now you try!

Chapter 6 · When People Get Hurt

A A four-line dialogue: somebody is hurt and you help.

 Hand in · W5

Complete. Then act it out.

A: Are you OK?

B: No. My _____ hurts.

A: Sit here. I can call the _____.

B: _____!

B Label the body.

Write: **head** · **arm** · **leg** · **back** on the picture.

C What's the problem?

Match.

1. headache

a.  head

2. stomachache

b.  stomach

3. backache

c.  back

D Emergency numbers quiz

Firefighters: 1 ___ ___ Ambulance: 1 ___ ___

E Who, where, what?

Look at the picture of Tomás. Say: “Tomás is hurt. He is in the playground.”

The Backpack



What is in the backpack?



water ($4 \times \frac{1}{2}$ litre)
agua embotellada



torch and batteries
linterna y pilas



whistle
silbato



tinned food
comida enlatada



masks
mascarillas



first-aid kit
botiquín



small radio
radio portátil



blanket
manta polar



biscuits and chocolate
galletas y chocolate



hand gel
gel antibacterial



My backpack is ready. Is yours?

After a big earthquake, there is no light. There is no water in the tap. The shops are closed. We need a special bag. It is the **emergency backpack**.

The real INDECI rules

24 h

For 1 day (24 hours).

8 kg

Not heavy: 8 kilos.

1 : 2

1 bag for 2 people.

1 L

Water: 1 litre for each person.

What is in the backpack?



water (4 × ½ litre)
agua embotellada



torch and batteries
linterna y pilas



whistle
silbato



tinned food
comida enlatada



masks
mascarillas



first-aid kit
botiquín



small radio
radio portátil



blanket
manta polar



biscuits and chocolate
galletas y chocolate



hand gel
gel antibacterial

We **must** have water. We **must** have a torch.

We **must not** put toys in the bag. It is heavy!

Why a Whistle?

A whistle is small. It is loud. People can hear you!

The Reserve Box

At home there is a big box too. It has food and water for more days.

Now you try!

Chapter 7 · The Backpack

A Your list of eight things. Hand in · W6

Draw 8 things for your backpack. Write the words.

1	2	3	4
5	6	7	8

B Yes or no?

Put ✓ or ✗: water ☐ · torch ☐ · TV ☐ · whistle ☐ · football ☐ · blanket ☐

C Count!

4 bottles of water. 2 cans. 1 torch. How many things? ____

D Nico's backpack

Nico's bag has: a TV, a teddy, water, crisps. What is wrong? Tell your friend.

E Take it home

Show this page to your family. Tick what you have at home.

CHAPTER 8

W7 · Producto y audiencia

The Safe Zone and the Drill



A drill at a school in Lima.

Photo: Joaquin Principe · CC BY-SA 4.0



Drop! Cover! Hold on!

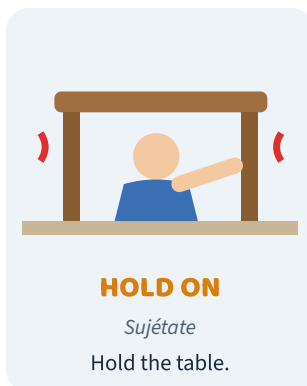
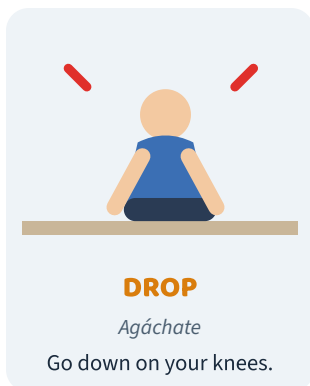
At school there are green signs. They say **ZONA SEGURA**. It means **safe zone**. In an earthquake, we go there.



A green sign: Zona Segura.

Photo: AgainErick · CC BY-SA 3.0 (also GFDL)

Drop, cover, hold on



Drop: go down on your knees. **Cover:** cover your head.
Hold on: hold the table.

The three moments of a Peruvian drill

The drill

- 1  The alarm rings. Drop, cover, hold on.
- 2  Walk out. Don't run. Don't shout. Don't push.
- 3  Go to the meeting point. The teacher counts.

Peru has a big drill for all the country. It is the **Simulacro Nacional**. Millions of children do it!

No corro, no grito, no empujo

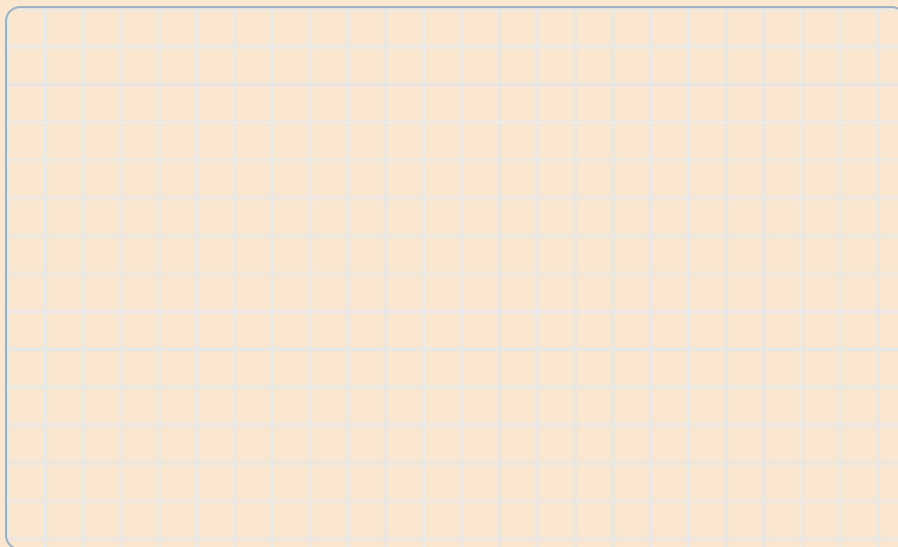
In Peru, children say: **No corro, no grito, no empujo.** I don't run. I don't shout. I don't push.

Now you try!

Chapter 8 · The Safe Zone and the Drill

A Draw your classroom. Hand in · W7

Draw your classroom from above. Draw the door. Colour the **safe zone** green.



B Put the drill in order.

☐ Walk out. ☐ Drop, cover, hold on. ☐ The teacher counts.

C Game: Simon says

Your teacher says “Simon says drop!” — drop! If the teacher doesn't say “Simon says”, don't move!

D Say the chant.

Drop, drop, drop! Cover, cover, cover! Hold on, hold on, hold on — until it stops!



Before, during or after?

Draw a line to **before, during** or **after**: Drop, cover, hold on · Pack the backpack · Go to the meeting point.

CHAPTER 9

W8 · Producto y audiencia

How a Safety Flyer Works



Signs are short and clear.

Photo: Gaboct · CC BY-SA 4.0



What's this? It's a flyer! Big words, big pictures.

A **flyer** is one page. It has a title, big pictures and short sentences. It tells people what to do.

Is your family ready?

BEFORE
Get ready!



Pack a backpack: water, a torch and a whistle. Put it by the door.

Our backpack is by the door.

DURING
Drop, cover, hold on!



Don't run. Go down. Cover your head. Hold on.

Stay under the table.

AFTER
Go out!



Walk to the meeting point. Don't run, don't shout, don't push. Near the sea? Go up!

Our meeting point is the park.

Emergency: 116 • 105 • 106 • Voice messages: 119

Model flyer — 70 words.

The parts of a flyer



Title

The big words at the top.



Heading

The name of each part.



Instruction

It says: do this!



Picture

It shows the action.



Caption

Small words under a picture.

The hidden instruction

“A whistle is very loud.” This means: **Pack a whistle!**

Now you try!

Chapter 9 · How a Safety Flyer Works

A Your three headings.

 Hand in · W8

Copy and complete:

BEFORE — Pack a _____.

DURING — Drop, cover, _____.

AFTER — Go to the _____.

B Label the model flyer.

Find on the flyer: the **title**, a **picture**, a **heading**. Point and say.

C Find the hidden instruction.

“The backpack is by the door.” → Put the backpack _____ the door.

D Good flyer or bad flyer?

Which is better? A: “DROP! COVER! HOLD ON!” B: “It is important that people think about what they can do.”

Writing My Flyer



Time to write!



First plan. Then write. Then draw!

Now you make your flyer. It has three parts: **BEFORE**, **DURING**, **AFTER**. Each part has a heading, an instruction and a picture.

BEFORE	DURING	AFTER
Heading:	Heading:	Heading:
Instruction:	Instruction:	Instruction:
Picture:	Picture:	Picture:

Word count: _____ (90–110)

Word bank

- Disasters** earthquake · tsunami · landslide · flood · volcano · ash
- Safety** safe zone · drill · exit · meeting point · shelter · danger · Drop, cover, hold on
- Backpack** emergency kit · water · torch · whistle · blanket · first-aid kit · radio
- Grammar** must · must not · before · during · after

Sentence starters

Pack a... · Keep your... · Drop, cover and... · Don't run! · Go to the... · Call...



LOOK!

Count your words! 90 is a lot — but the frames help you.

Now you try!

Chapter 10 · Writing My Flyer

A Your first draft, with its three drawings. Hand in · W9

Complete the flyer frame. Then draw one picture in each box.

BEFORE — Get ready!

Pack a backpack. Put in water, a _____, a
_____ and a _____. Keep it by the
_____.

draw

DURING — Drop, cover, hold on!

Don't _____. Go down on your _____. Cover
your _____. Hold on to a _____.

draw

AFTER — Go out!

Walk to the _____ point. Don't run, don't shout,
don't _____. Near the sea? Go _____!

draw

B must or must not?

You _____ pack water. You _____ run on the stairs.

CHAPTER 11

W10 · Producto y audiencia

Making It Better



We check our flyers together.



Oops! My flyer has mistakes. Can you help me?

Swap flyers with a friend. Read it. Use the checklist. Say two good things and one thing to change.

✓ Checklist

- ☐ My flyer has a BEFORE part.
- ☐ My flyer has a DURING part.
- ☐ My flyer has an AFTER part.
- ☐ I use must, must not or should.
- ☐ I say what the family must have ready.
- ☐ I say where to go.

Check the verbs

Instructions start with a verb: **Pack** a torch. **Go** up. **Don't** run.



Nico's flyer — find the mistakes!

- BEFORE: Pack a backpack.
- DURING: Run outside!
- AFTER: Go to the beach.
- Cover your feet.

Read it out loud

Read your flyer to the class. Speak loud. Speak slowly.
Look at the people.

Now you try!

Chapter 11 · Making It Better

A

Your finished flyer.



Hand in · W10

Copy your flyer on a big paper. Use colours. Draw three pictures.

B

Two stars and a wish



I like...



I like...



Change...

C

Fix the flyer.

Find 3 mistakes in Nico's flyer (in the box above).

CHAPTER 12

W11 · Producto y audiencia

The Safety Fair



The safety fair at school.



We did it! Our flyers are ready!

Today is the safety fair. Families come to school. You show your flyer. You say what to do.



Presenting my flyer

1. Hello! My name is ... This is my flyer.
2. Before an earthquake, ...
3. During an earthquake, ...
4. After an earthquake, ...
5. Thank you! Any questions?

And in my house?

Look at your home. Do you have a backpack? Do you have water? Write one thing you can do.

At home	We have it ✓	We don't have it ✗
emergency backpack	☐	☐
water	☐	☐
torch and batteries	☐	☐
whistle	☐	☐
first-aid kit	☐	☐
a family meeting point	☐	☐

Remember the big question? Now you know the answer.
You are ready!

Now you try!

Chapter 12 · The Safety Fair

A Your presentation and the family reflection.

 Hand in · W11

Say: “Hello! This is my flyer. Before an earthquake... During... After...”

Then write about your home:

We have _____. We don't have _____. I can _____.

B Questions visitors might ask

Practise: “What's in the backpack?” — “Water, a torch...”

C How did I do? (Rubric)

Colour the stars: ★★★★★ I know the words · ★★★★★ I can say what to do · ★★★★★ I made my flyer

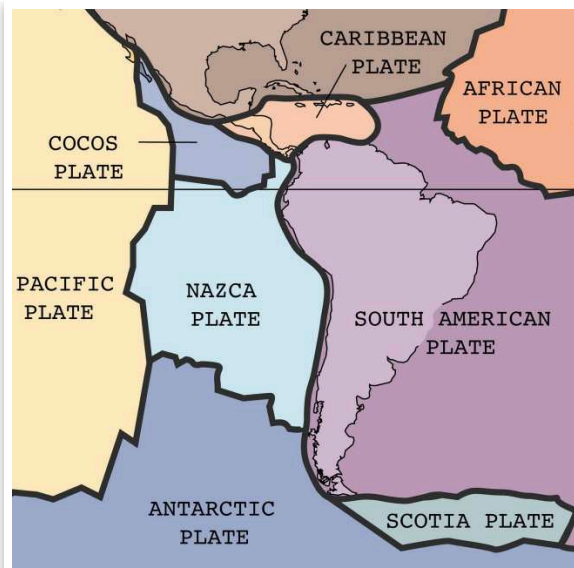
Peru in Photos

Real photos of earthquakes and volcanoes in Peru.



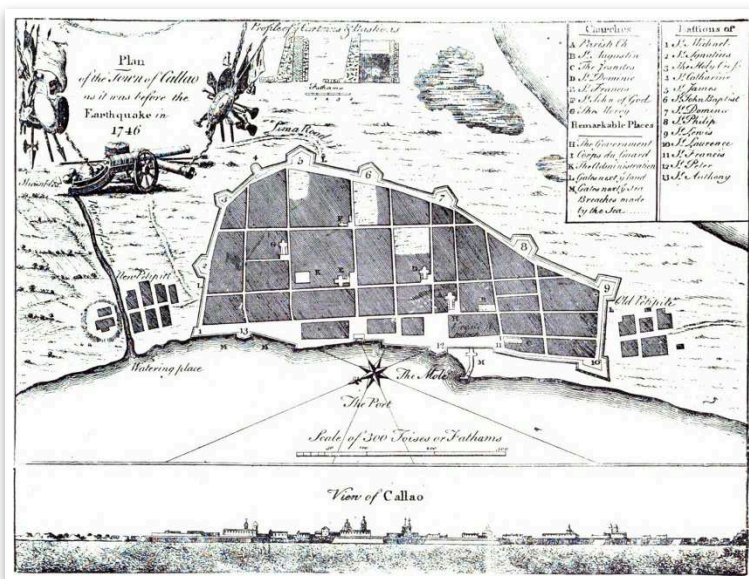
Many old houses in Lima have wooden balconies.

Photo: Felipe Restrepo Acosta · CC BY-SA 4.0



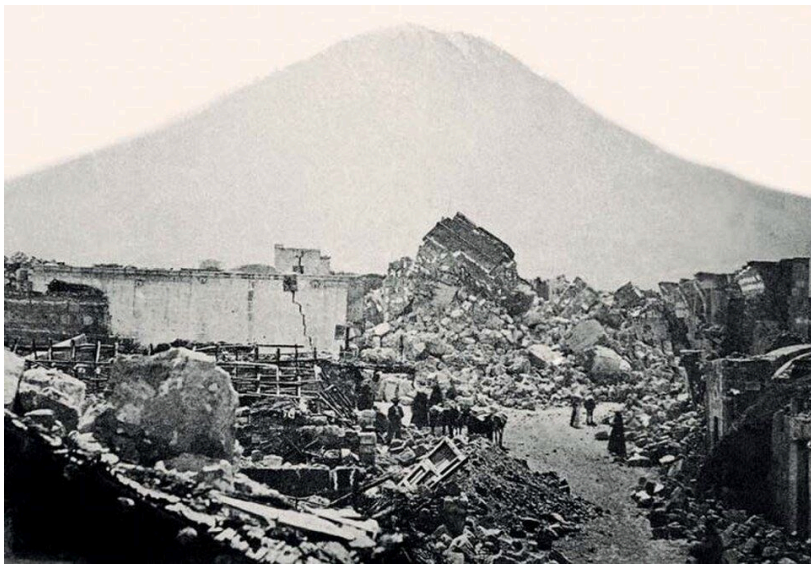
Peru is where the Nazca Plate and the South American Plate meet.

Photo: U.S. Geological Survey · Public domain (USGS)



This old map shows the port of Callao before the earthquake and tsunami of 1746.

Photo: Unknown 18th-century engraver; reproduced by The West Coast Leader, Ltd. · Public domain



The earthquake of 1868 knocked down many buildings in Arequipa.

Photo: Unknown photographer, c. 1868 · Public domain



The tsunami of 1868 carried the ship Wateree onto the land at Arica.

Photo: Unknown photographer, 1868; Henry Clay Cochrane Collection, USMC Archives · CC BY 2.0



Parts of the Wateree are still on the beach near Arica today.

Photo: Pinot · Public domain (released by the author)



The Morro de Arica is a tall rock hill next to the sea.

Photo: Jowyto · CC BY-SA 3.0



Huascarán is the highest mountain in Peru.

Photo: Maurice Chédel · Public domain (released by the author)



From the sky you can see the path of the 1970 avalanche from Huascarán down to Yungay.

Photo: George Plafker, U.S. Geological Survey · Public domain (USGS / NOAA)



After the avalanche of 1970, the statue on the cemetery hill in Yungay was still standing.

Photo: U.S. Geological Survey · Public domain (USGS)



In 1980, people built this gate at the Yungay memorial. Huascarán is behind it.

Photo: DB · Public domain (released by the author)



Today the white statue in Yungay looks out at Huascarán.

Photo: Jgaray2000 · CC BY-SA 4.0



These palm trees grew in the main square of old Yungay. They are still there today.

Photo: Yhhue91 · CC BY-SA 4.0



Astronauts took this photo of Camaná, where a tsunami came on land in 2001.

Photo: NASA Earth Sciences and Image Analysis Laboratory, Johnson Space Center · Public domain (NASA)



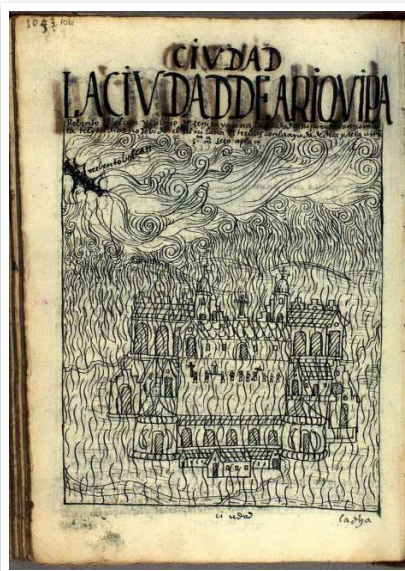
After the earthquake of 2007, broken bricks filled the streets of Pisco.

Photo: Senior Airman Shaun Emery, U.S. Air Force · Public domain (U.S. military)



Helpers unloaded boxes of supplies for families after the 2007 earthquake.

Photo: Tech. Sgt. Jeremy Lock, U.S. Air Force · Public domain (U.S. military)



This old drawing shows ash falling on Arequipa when Huaynaputina erupted in 1600.

Photo: Felipe Guaman Poma de Ayala · Public domain



El Misti volcano stands over the city of Arequipa.

Photo: Josep M. Gracia · CC BY-SA 4.0



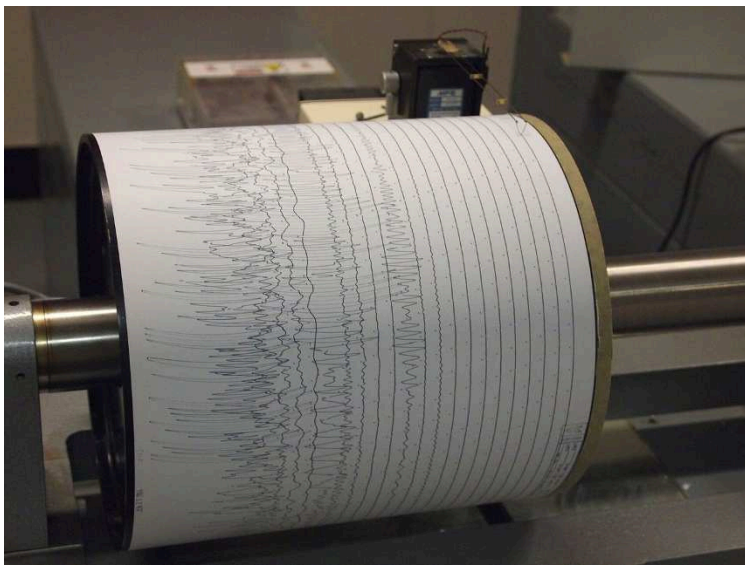
Sabancaya volcano sent up a big cloud of ash in 2017.

Photo: Ministerio de Defensa del Perú · CC BY 2.0



In 2006, Ubinas volcano blew out a cloud of ash. This photo was taken from space.

Photo: NASA astronaut aboard the International Space Station · Public domain (NASA)



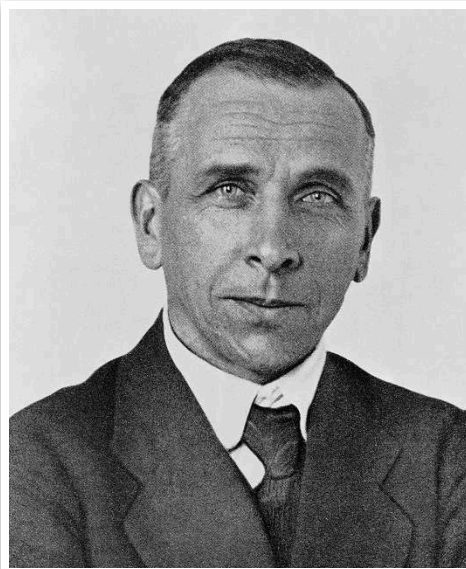
A seismograph draws wiggly lines when the ground shakes.

Photo: Z22 · CC BY-SA 3.0



Charles Richter made a scale to tell how strong an earthquake is.

Photo: Gil Cooper, Los Angeles Times · CC BY 4.0



Alfred Wegener said that the continents move very slowly.

Photo: Unknown photographer · Public domain



In Peru, this sign shows a safe place to go in an earthquake.

Photo: AgainErick · CC BY-SA 3.0 (also GFDL)



This sign in Lima shows which way to go if a tsunami comes.

Photo: Gaboct · CC BY-SA 4.0



Students in Lima practise an earthquake drill.

Photo: Joaquin Principe · CC BY-SA 4.0

Timeline of Peru's Earthquakes, Tsunamis and Volcanoes

- 1582** —An earthquake breaks Arequipa.
- 1600** —The volcano Huaynaputina explodes.
- 1604** —An earthquake and a tsunami in the south.
- 1687** —An earthquake in Lima.
- 1746** —A big earthquake and tsunami in Lima and Callao.
- 1868** —A tsunami puts a ship on the sand in Arica.
- 1922** —The IGP starts in Huancayo.
- 1940** —An earthquake in Lima.
- 1970** —The Áncash earthquake. An avalanche covers Yungay.
- 1972** —Peru makes INDECI.
- 1974** —An earthquake in Lima and Callao.
- 2001** —An earthquake in Arequipa. A tsunami in Camaná.
- 2007** —The Pisco earthquake.
- 2016** —Sabancaya volcano starts to erupt.
- 2019** —A big earthquake in the Amazon.
- 2025** —The IGP writes about 840 earthquake reports.
- 2026** —Millions of children do the national drill.

Timeline of the World

- 1556** —China: the worst earthquake ever.
- 1755** —Portugal: an earthquake and a tsunami in Lisbon.
- 1906** —USA: an earthquake in San Francisco.
- 1912** —Alfred Wegener says: the land moves!
- 1935** —Charles Richter makes his scale.
- 1960** —Chile: the biggest earthquake ever measured.
- 1964** —USA: a big earthquake in Alaska.
- 2004** —Indonesia: a very big tsunami.
- 2010** —Chile: a big earthquake.
- 2011** —Japan: an earthquake and a tsunami.
- 2015** —Nepal: an earthquake near Mount Everest.
- 2023** —Türkiye and Syria: a big earthquake.
- 2025** —Russia: a big earthquake. Waves cross the Pacific.

Glossary

The words of this unit, with Spanish.

after (*después*) — at a later time: after the earthquake

aftershock (*réplica*) — a smaller earthquake that comes after a big one

ankle (*tobillo*) — the part of the body between your leg and your foot

ash (*ceniza*) — the grey powder that comes out of a volcano

avalanche (*avalancha / aluvión*) — a very large amount of ice, snow and rock falling fast down a mountain

back (*espalda*) — the part of your body behind you, from your neck to your bottom

backache (*dolor de espalda*) — a pain in your back

before (*antes*) — at an earlier time: before the earthquake

blanket (*manta*) — a warm cover for your bed or for sleeping outside

bruise (*moretón*) — a blue or purple mark on your skin after a knock

cut (*corte*) — an opening in your skin that bleeds

danger (*peligro*) — something that can hurt you

drill (*simulacro*) — practice for what to do in an emergency

Drop, cover, hold on (*Agáchate, cúbrete, sujétate*) — what you do during the shaking

during (*durante*) — while something is happening

earthquake (*sismo / temblor / terremoto*) — a sudden shaking of the ground, big or small: in English it is always “earthquake”

emergency kit (*mochila para emergencias*) — a bag with the things you need in an emergency

epicentre (*epicentro*) — the place on the ground right above where an earthquake starts

exit (*salida*) — the way out of a building

fault (*falla*) — a long crack in the rock where the ground can suddenly move

first-aid kit (*botiquín*) — a box with plasters, bandages and things to help people who are hurt

flood (*inundación*) — a lot of water covering land that is usually dry

have to (*tener que*) — it is necessary: we have to leave now

head (*cabeza*) — the top part of your body, with your eyes and mouth

headache (*dolor de cabeza*) — a pain in your head

hurt (*doler / lastimarse*) — to feel pain: my knee hurts

intensity (*intensidad*) — how strongly the shaking is felt in one place

knee (*rodilla*) — the middle of your leg, where it bends

landslide (*deslizamiento / huaico*) — rocks and earth sliding down a hill

lava (*lava*) — hot liquid rock that comes out of a volcano

magnitude (*magnitud*) — a number that shows how strong an earthquake is

meeting point (*punto de reunión*) — the safe place outside where everyone meets after leaving

must / must not (*debe / no debe*) — for rules: you must pack water; you must not run

plate (*placa tectónica*) — a giant piece of the Earth's outside

Ring of Fire (*Cinturón de Fuego*) — the line of coasts around the Pacific with many earthquakes and volcanoes

safe zone (*zona segura*) — the safest place to be in an earthquake

seismograph (*sismógrafo*) — a machine that measures the shaking of the ground

shelter (*refugio / albergue*) — a safe place to stay

should (*debería*) — for advice: you should check your backpack

stomachache (*dolor de estómago*) — a pain in your stomach

torch (*linterna*) — a small light you carry in your hand

tsunami (*tsunami / maremoto*) — very big waves caused by an earthquake under the sea

volcano (*volcán*) — a mountain where hot rock, gas and ash come out

whistle (*silbato*) — a small thing you blow into to make a loud sound

How am I doing?

The unit rubric. AD is the highest level.

I can...	AD	A	B	C
I name natural disasters, safety words, the things in the emergency kit and the parts of the body.	I use them and I explain what one of them means to somebody who does not know it.	I use the words on my own, in sentences, when I am talking about safety.	I say most of the words when I see the picture.	I say a few of the words with help.
I read a table and a chart about earthquakes in Peru and I say what they show.	I notice something in the data that nobody pointed out to me.	I ask and answer questions about the data and I compare two earthquakes.	I find information when I am asked a direct question.	I find one piece of information if somebody shows me the column.
I tell what happened in a real Peruvian earthquake using the past simple.	I tell it in order and I add what happened afterwards.	I tell what happened in four sentences, in order and in the past.	I write two or three sentences in the past with some mistakes.	I copy sentences in the past.
I use must, must not and should to give safety rules and advice.	I explain to somebody else why one is a rule and the other is advice.	I choose must, must not or should correctly and I give the reason.	I write rules but I mix up must and should.	I copy a rule that is given to me.
I organise safety instructions into before, during and after.	I notice when an instruction could go in two moments and I say why.	I sort them all and I use the three moments to organise my own flyer.	I sort most of the actions with a bit of help.	I put two of the three moments in the right place.
I write a safety flyer of 90 to 110 words and I present it to other people.	My flyer works for somebody who is younger than me, and I prove it.	I write the three parts, with a heading, an instruction and a drawing each, and I present it.	I write the three parts but they are short or unclear.	I write some sentences and I show my drawing.

Sources and Credits

Sources

- Instituto Geofísico del Perú (IGP) — earthquake reports and catalogue, igp.gob.pe
- INDECI — Combo de la Supervivencia; tsunami recommendations, gob.pe/indeci
- Dirección de Hidrografía y Navegación (DHN) — tsunami warning service
- USGS Earthquake Hazards Program — event pages and “Cool earthquake facts”
- NOAA NCEI — Global Historical Tsunami and Significant Earthquake databases
- USGS Circular 639 (1970): The Peru earthquake of May 31, 1970
- US Navy reports from USS Wateree and USS Powhatan, Arica 1868 (Naval History and Heritage Command)
- Smithsonian Global Volcanism Program — Huaynaputina, El Misti, Ubinas, Sabancaya
- Britannica — Charles F. Richter; Alfred Wegener; Ancash earthquake of 1970
- Minedu / INDECI — Simulacro Nacional Multipeligro 2026
- Nordic International School — Teacher guide, Grade 3, Unit 5: When the Earth Moves

Photo credits

- yungay-1970-avalanche-aerial.jpg — George Plafker, U.S. Geological Survey (NOAA NCEI Natural Hazards Image Database, image 785), Public domain (USGS / NOAA). <https://www.ngdc.noaa.gov/hazardimages/#/all/7/image/785>
- yungay-christ-statue-1970.jpg — U.S. Geological Survey, Public domain (USGS). https://commons.wikimedia.org/wiki/File:Statue_of_Christ_at_Cemetery_Hill_overlooking_old_Yungay.jpg
- yungay-1980.jpg — DB (family archive), Public domain (released by the author). https://commons.wikimedia.org/wiki/File:Yungay,_1980.jpg
- yungay-palms.jpg — Yhhue91, CC BY-SA 4.0. https://commons.wikimedia.org/wiki/File:Pueblo_de_Yungay_en_%C3%81ncash_Per%C3%BA.jpg
- yungay-christ-statue-today.jpg — Jgaray2000, CC BY-SA 4.0. [https://commons.wikimedia.org/wiki/File:Cristo_de_Yungay_en_el_cementerio_de_Yungay_\(Ancash,_Per%C3%BA\).jpg](https://commons.wikimedia.org/wiki/File:Cristo_de_Yungay_en_el_cementerio_de_Yungay_(Ancash,_Per%C3%BA).jpg)
- huascaran-peaks.jpg — Maurice Chédel, Public domain (released by the author). https://commons.wikimedia.org/wiki/File:Nevado_Huascaran.jpg
- callao-1746-plan.jpg — Unknown 18th-century engraver; reproduced by The West Coast Leader, Ltd. (1925), Public domain. [https://commons.wikimedia.org/wiki/File:\(CCC_No._10._1925\)-_Plano_y_vista_del_Callao,_tal_como_en_el_a%C3%B1o_1746_cuando_fu%C3%A9_destruido_por_el_terremoto.png](https://commons.wikimedia.org/wiki/File:(CCC_No._10._1925)-_Plano_y_vista_del_Callao,_tal_como_en_el_a%C3%B1o_1746_cuando_fu%C3%A9_destruido_por_el_terremoto.png)
- uss-wateree-arica-1868.jpg — Unknown photographer, 1868; Henry Clay Cochrane Collection, USMC Archives (Quantico, USA), CC BY 2.0. [https://commons.wikimedia.org/wiki/File:The_U.S.S._Wateree_\(14037445157\).jpg](https://commons.wikimedia.org/wiki/File:The_U.S.S._Wateree_(14037445157).jpg)
- wateree-boilers-arica.jpg — Pinot (English Wikipedia user), Public domain (released by the author). https://commons.wikimedia.org/wiki/File:Chile_2006_022.jpg
- morro-arica.jpg — Jowyto, CC BY-SA 3.0. https://commons.wikimedia.org/wiki/File:Morro_de_Arica_desde_la_playa.JPG
- arequipa-1868.jpg — Unknown photographer, c. 1868 (Biblioteca Nacional del Perú), Public domain. https://commons.wikimedia.org/wiki/File:Arequipa_devastada_por_el_terremoto_de_1868.jpg
- camana-coast-from-space.jpg — NASA Earth Sciences and Image Analysis Laboratory, Johnson Space Center, Public domain (NASA). [https://commons.wikimedia.org/wiki/File:Tsunami_Vulnerability_in_Camana,_Peru_\(2282\).jpg](https://commons.wikimedia.org/wiki/File:Tsunami_Vulnerability_in_Camana,_Peru_(2282).jpg)
- pisco-2007-after.jpg — Senior Airman Shaun Emery, U.S. Air Force, Public domain (U.S. military). https://commons.wikimedia.org/wiki/File:Pisco_after_the_2007_quake.jpg
- pisco-2007-aid-lima.jpg — Tech. Sgt. Jeremy Lock, U.S. Air Force (via U.S. Army), Public domain (U.S. military). https://commons.wikimedia.org/wiki/File:Aid_after_earthquake_in_Peru.jpg
- misti-arequipa.jpg — Josep M. Gracia, CC BY-SA 4.0. https://commons.wikimedia.org/wiki/File:ARE_-_El_Misti_volcano_above_Arequipa,_Peru,_2013.jpg
- ubinas-eruption-from-space.jpg — NASA astronaut aboard the International Space Station (NASA Earth Observatory), Public domain (NASA). https://commons.wikimedia.org/wiki/File:Ubinas_ash_cloud_-_ISS.jpg

- sabancaya-eruption.jpg — Ministerio de Defensa del Perú (Flickr gallery), CC BY 2.0.
https://commons.wikimedia.org/wiki/File:Erupci%C3%B3n_en_el_volc%C3%A1n_Sabancaya,_Per%C3%BA.jpg
- huaynaputina-arequipa-1600.jpg — Felipe Guaman Poma de Ayala, Public domain.
https://commons.wikimedia.org/wiki/File:Arequipa_Ascheregen_Huaynaputina_1600.jpg
- plates-map-south-america.jpg — U.S. Geological Survey (cropped by Beyond My Ken), Public domain (USGS).
https://commons.wikimedia.org/wiki/File:South_American_plates.png
- wegener-portrait.jpg — Unknown photographer, Public domain.
https://commons.wikimedia.org/wiki/File:Alfred_Wegener_ca.1924-30.jpg
- charles-richter.jpg — Gil Cooper, Los Angeles Times (UCLA Library Special Collections), CC BY 4.0.
https://commons.wikimedia.org/wiki/File:Charles_Richter_analyzing_a_seismograph_log.jpg
- seismograph-drum.jpg — Z22, CC BY-SA 3.0.
https://commons.wikimedia.org/wiki/File:Seismogram_at_Weston_Observatory.JPG
- zona-segura-sign.jpg — AgainErick, CC BY-SA 3.0 (also GFDL).
https://commons.wikimedia.org/wiki/File:Sign_Peru_Zona_Segura_en_casos_de_sismo.jpg
- tsunami-evacuation-sign-lima.jpg — Gaboct, CC BY-SA 4.0.
https://commons.wikimedia.org/wiki/File:Letrero_sobre_peligro_de_tsunami_en_la_quebrada_de_Armendariz.jpg
- school-drill-lima.jpg — Joaquin Principe, CC BY-SA 4.0.
https://commons.wikimedia.org/wiki/File:Simulacro_nacional_multipeligro_Per%C3%BA_31_mayo_2022_-2.jpg
- lima-balconies.jpg — Felipe Restrepo Acosta, CC BY-SA 4.0. https://commons.wikimedia.org/wiki/File:2017_Lima_-_Balcones_en_el_273_jir%C3%B3n_Ca%C3%B1ete.jpg

Illustrations: 3D art made with Canva AI for NIS. Characters: Fun for Nordic © Nordic International School. 3D models: made for this book.

When the Earth Moves

Peru shakes! Read about five big earthquakes, giant waves and volcanoes. See real photos. Learn what to do. Then make a safety flyer for your family.

Inside this book

- 12 chapters, one for each week
- Activities after every chapter
- Peru in Photos
- Two timelines
- Glossary with Spanish
- How am I doing? rubric



Starters edition • Pre-A1

Nordic International School • Grade 3 • Unit 5 • Lima, Peru